Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type Mammalian, adherent, suspension

Species Used Human, B lymphomas: BJAB, P3HR-1, B95-8; Rat-1

Molecules Electroporated DNA: ccc *Bam* Z (5 kB), *Eco* A cosmid > 40 kB, *Sal* A cosmid > 40 kB

Before the Pulse

Cell Growth Medium RPMI-10 (GIBCO/BRL, Sigma)

Growth Phase at Harvest Log

Pre-pulse Incubation 10 min.

Wash Solution None

The Pulse

Instruments Used Gene Pulser® apparatus, Capacitance

Electroporation Temperature 4°C

Electroporation Medium* RPMI-10 + 15% serum

Cell Density 5 x 10 (6) or 10 (7) per 350 µl

Volume of Cells 350 µl

DNA Concentration 10 to 50 µg / cuvette

DNA Resuspension Buffer RPMI-10 or Phosphate Buffered Saline (PBS) or water

Volume of DNA 10 µl

Cuvette Gap 0.4 cm

Voltage 0.20 to 0.25 kV

Field Strength 0.08 to 0.1 kV/cm

Capacitor 960 µF

Resistor (Pulse Controller) Ω none

Time Constant 28 to 40 msec

After the Pulse

Outgrowth Medium Not given

Outgrowth Temperature 37°C

Length of Incubation Variable

Selection Method or Assay Used Protein expression by immunoblot, immunofluorescence, generation of recombinants.

Electroporation Efficiency > 20%

Per Cent Survival Variable %

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.
PNAS **88**:1546-1550 (1991).
PBS: 1x = 8g NaCl,0.2g KCl,0.2g KH2PO4, 1.15g Na2HPO4

Name of Submittor Sankar Swaminathan, M.D.

Institution Address Harvard Medical School
Brigham & Women's Hospital
Dept of Infectious Diseases
20 Shattuck St.
Boston, MA 02115

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